

Bui Hai

Viet Nam

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Python | Machine Learning | CI/CD | LLM | RAG | Azure SDK | Docker | Lambda | Open-CV | Torch | Keras | C/C++ | Systems | AWS | SQL | Kafka | Unit Test | OOP

PERSONAL EXPERIENCE

Cinnamon AI - *AI Engineer.*

Feb 2020 - Present

- **Flax Scanner Hub**

- Built an intelligent Document Information Extraction platform leveraging **Azure Document Intelligence**, **LLMs**, and **RAG** techniques, achieving **~89%** extraction accuracy across diverse document types.
- Designed a scalable **microservices** architecture capable of serving **1M** concurrent users, ensuring **fair resource distribution** across tenants and processing over **10,000 pages per hour** using 8 distributed services.
- Built an LLM Gateway, centralizing third-party API calls - enabling token usage tracking for client billing. Implemented response caching and latency simulation, saving cost for each API call in performance testing.
- Designed and implemented a document abstraction layer that provides a **consistent** interface for extracting content, layout, and structural metadata from diverse document formats.
- Designed and implemented a **multi-document** question-answering system leveraging **RAG** pipelines and **LLMs**, enabling users to retrieve, analyze, and interact with information through natural language conversations.
- Designed and implemented an **AI evaluation** platform that performs automated accuracy benchmarking, release-to-release comparisons, and failure analysis, improving visibility into model quality and performance trends.

- **Document Information Extraction**

- Develop AI model for a Document Information Extraction system, ensuring customized output to fit client-specific requirements, success delivery for **10+ clients**.
- Designed and developed a graph-based rule engine that models project-specific business logic as configurable dependency graphs, reducing developer workload by **70%** while enabling rule updates without application rebuilds.
- Conducted model research, evaluation, and performance benchmarking to select the most effective architectures for document information extraction across diverse document types.

- **Engineering Excellence / Internal Platform**

- Develop Automated AWS Setup: Built a deployment **automation** tool for AWS (EC2, S3, ECR), reducing setup time for new client projects by **90%**.
- Standardizing working flow of AI Engineers: develop **CI/CD** for internal AI library and AI pipeline.
- Develop Image Labeling System: a bounding box annotation tool, streamlining dataset creation for AI training.
- Technical Leadership: Served as **Project Technical Lead**, defining AI solutions, mitigating risks, and communicating strategies with clients and PMs.
- Develop Multi-Format Image Reader: Developed a reader supporting PDF, PNG, JPG, HEIC, improving usability and reducing development time by **10%** per project.
- Develop Project Performance Tracking: Built an automated tracking system that monitors project performance (speed, milestones, logging hours), saving **10%** of management time.
- Hugging Face Integration: Extended AI capabilities to Hugging Face, enabling non-technical users to utilize AI functions, reducing developer dependency by **90%**.

- **Video Watching Recommend**
 - Developed a video recommendation system for the **Lotus social network** using **deeplearning** for **video classification** and **personalized recommendations**.
 - Integrated AI models into the Lotus system, handling over **1,000 daily users**, utilizing Python, Torch, OpenCV, Kafka, and MySQL.

- **OCR Handwriting**
 - Implemented deep learning models (CNN, RNN) for **handwriting OCR**.
 - Built a data pipeline to generate up to **60 million** training images for **Vietnamese Handwriting OCR** models.
 - Developed a data transfer protocol between servers and cameras.
- **Face Detection**
 - Integrated deep learning models for **face recognition**

EDUCATION

Hanoi University of Industry - Bachelor of Computer Science	Aug 2017 - May 2021
Rubik AI Talent -Certificate of Computer Science	Apr 2019 - Sep 2019
FUNiX University -Certificate of Automotive Programming Scholarship	May 2019 - Nov 2020

Honors & Awards

• 1st of Scientific research competition in Hanoi university of Industry: Auto Detecting students cheating in exam room.	2021
• National semi-finals "FPT - Digital Race": Self-driving car programming contest.	2020
• 1th of the Bedrock Hackathon: Automatically fill information to many different forms.	2019
• 7th of the northern semi-final "FPT - Digital Race": Self-driving car programming contest.	2019
• 5th of CINNAMON AI MARATHON: Develop Handwriting OCR for Vietnamese Address.	2019